

TECHNICAL SPECIFICATION FOR INDOOR RESIN CAST TYPE 11 KV CURRENT & POTENTIAL TRANSFORMER HOUSED IN A COMMON UNIT SUITABLE FOR 11 KV BULK CONSUMER

1. SCOPE :

This specification covers the design, manufacture, assembly, testing at the manufacturer's works, supply & delivery at WBSEDCL's stores/sites anywhere in WB including unloading of indoor current & potential transformers of 11KV voltage class as specified in Schedule - A for metering services suitable for 11 KV Bulk Consumer.

2. STANDARD :

The indoor current & potential transformers unit and accessories covered by this specification shall comply with the requirement of the latest edition of the following standards unless otherwise stated in this specification.

IS:2705 & IS 3156 : Specification for Instrument Transformers.

3. DEVIATION :

Normally the offer should be as per Technical Specification without any deviation. But any deviation felt necessary to improve performance, efficiency and utility of equipment must be mentioned in the Deviation Schedule with reasons duly supported by documentary evidence. Such deviations suggested may or may not be accepted by the Company. Deviations not mentioned in Deviation schedule will not be considered.

4. DESIGN & CONSTRUCTION OF INDOOR 11 KV CURRENT & POTENTIAL TRANSFORMERS:

The design features and construction details of indoor current & potential transformer shall be in accordance with the requirement stipulated in Schedule - A :

- i) The current & potential transformers shall be complete in all respects and shall conform to the modern practice of design and manufacture.
- ii) The single phase current & potential transformers shall be of low Reactance indoor resin cast type, 50 Hz, self cooled covered in a suitable housing both in same alignment (as per drawing).
- iii) The maximum permissible temperature rise of the current & potential transformer winding when carrying a primary current equal to the rated continuous current at rated frequency and with rated burden over an ambient temperature shall not exceed 55°C.
- iv) The current and potential transformer shall be Resin cast type. The current & potential transformers unit shall be suitable for placed in a common housing in separate compartment with suitable support (as per drawing).
- v) The CT & PT Unit shall be complete with all accessories with indoor arrangement, grounding terminals and name plate. Arrangement for PT primary & secondary neutral terminal earthing shall be provided.
- vi) The CT & PT Unit shall be resin cast type provided with class - E insulation. The design and construction of CT & PT Unit shall be sufficient to withstand the thermal and mechanical stresses resulting from the specified short circuit currents and specified duration as mentioned in General Technical Specification.
- vii) The core of the CT & PT Unit shall be high grade non-ageing, electrical, silicon laminated steel of low hysteresis loss and high permeability to ensure high accuracy at both normal and over current.

- viii) The exciting current shall be as low as possible and the CT & PT Unit shall be capable of maintaining its rated accuracy at different burdens and within saturation limits.
- ix) Rating plate marking of both CT & PT shall be provided as per relevant clause of IS:2705& IS 3156.
- x) The CT & PT Unit Characteristics shall be such as to provide satisfactory performance for burdens ranging from at least 25% to 100% of rated burden in case of metering CT & PT Unit cores.
- xi) When operating the current transformer between 5% of the rated current and the rated continuous thermal current, the voltage transformer shall not exceed the limits of voltage error and phase displacement corresponding to its class within the specified range of burden and between 80% and 120% of the rated voltage.
- xii) When operating the voltage transformer between 80% of the rated voltage and the rated voltage multiplied by the rated voltage factor, the current transformer shall not exceed the limits of current error and phase displacement within the range of current corresponding to its class and within the specified range of burden.
- xiii) The CT & PT Unit secondary to be used for bulk metering and instruments shall be of accuracy class and I S F as specified. The saturation factor of this core shall be low enough so as not cause any damage to measuring instruments in the event of maximum short circuit current.
- xiv) The ratio changing arrangement shall be provided on secondary side of the CT.

5. GROUNDING TERMINALS :

Two grounding terminals on diagonally opposite sides of common housing with adequate size suitable for connecting M.S.Flat of size 50 mm x 6 mm shall be provided.

6. COMMON HOUSING UNIT :

- i) CT, PT, Meter& TTB (meter not in scope of bidder, but proper space to be provided in upper compartment as per drawing) should be placed in a common housing with proper separation in between them and adjustable gland plate at bottom suitable for 11 KV 3C 185 or 3C 95 sqmm XLPE cable.
- ii) Suitable fixing support for 11 KV cable both incoming & outgoing should be provided in side the common housing.
- iii) All 2.5 sqmm for PT & 4 sqmm for CT wire 1.1 kv grade PVCFRLS copper cable from secondary side of CT & PT should be placed and pass through a duct from one chamber to another chamber.
- iv) Riser for connecting CT & PT will be of size having 25X4 mm flat Copper bar.
- v) All secondary wires should be coloured mark as RED Phase : Red, Yellow phase : Yellow, Blue phase : Blue , Neutral : Black & Earth : Green. Both end of wires with same stated marking. Plastic ferrule should be conforming to IS.
- vi) Proper Phase to Phase and Phase to Ground clearance to be maintained as per Electric Rule and applicable standard.
- vii) Common Housing shall be provided with Degree of Protection of IP : 5X.
- viii) Proper Lifting hooks to be provided for Common housing.
- ix) For both CT and PT chamber, proper access of both unit to be provided from front and rear side for maintenance purpose
- x) Proper sealing arrangement to be provided for all chamber (separately for CT, PT and meter compartment).
- xi) All secondary wires emanating from CT & PT will be housed in the upper compartment with sufficient length so that it can be connected with energy meter installed in upper portion of the chamber.
- xii) Cable entry shall be at Primary side (P1) side of CT
- xiii) Thickness of steel sheet may be 2 mm
- xiv) All HV metal part shall be adequately insulated with fire retardant non hygroscopic anti tracking sleeve for rated voltage.
- xv) Door hinge of metal chamber shall be concealed type.
- xvi) In CT PT chamber, proper position of welded net including louvers shall be provided in left side of the unit as per drawing.

7. PAINTING :

- i) The tank and top metallic cover shall be painted with epoxy paint. All steel surfaces shall be cleaned by sand blasting or chemical process as required to produce a smooth surface, free of scale, grease and dirt.
- ii) Paints shall be carefully selected to withstand tropical heat, rain etc. The paints shall not scale off or crinkle or be removed by abrasion due to normal handling.
- iii) Painting shade of common housing shall be RAL 7032 for external & internal surface

8. TESTS AT FACTORY AND TEST CERTIFICATES :

- i) Each C.T&PT shall comply with the requirements of routine test as specified in the relevant Clause of IS:2705& IS 3156.
- ii) Routine test at manufacturer's works shall be carried out in presence of representative of WBSEDCL as per relevant IS.
- iii) All Acceptance tests shall be carried out at the manufacturer's works on every lot offered for inspection as per relevant IS. In addition to above, all routine tests are also to be carried out on CT & PT Unit as per relevant IS. Selection of samples for acceptance test as well as rejection and retesting shall be guided by relevant IS.

11. TEST REPORTS AND TYPE TESTS :

The Bidder should submit all the type test report on 11KV CT & PT (one no. each) with CTR:-100-50/ 5A and 50--25 /5A ratio separately as per IS:2705 & IS 3156 including IP 5X test report for common housing from CPRI/NABL accredited/Govt. recognized Test House or Laboratory carried out within 5 years from the due date of submission of bid having identical design, rating and type as that of the tendered item as pre-requisites along with the bid failing which their offer may not be technically acceptable.

The Type Test certificates of the NABL accredited/govt. recognized test house or laboratory should however, bear the logo of NABL accreditation.

Temperature Rise test & Degree of Protection test report on whole unit (after installation of both CT & PT) shall be provided by the bidder.

Each Type Test Report shall comply the following information with test results.

- i) Complete identification, date and serial no.
- ii) Method of application where applied, duration and interpretation of each test.

12. TENDER DRAWING, CATALOGUE AND TEST REPORTS :

One copy of the following drawings and catalogue shall be submitted with each copy of tender for evaluation :

- i) General arrangement drawings showing front elevation, side view, plan along with all accessories, mounting arrangement, creepage distance of the bushing, electrical diagram of primary and secondary connection with polarity marking, terminal arrangement of secondary terminal box, size of primary terminals, grounding terminals and lifting lugs, net and shipping weight, dimension etc.
- ii) Name and rating plate diagram.
- iii) Clear drawing of CT & PT arrangement, mentioning details dimension as per technical specification, to be submitted along with the bid. All type tests to be done on the items which will be designed exactly on these drawing and the certified copy of reports of same to be submitted.

13. CONTRACT DRAWINGS AND MANUALS:

13.1 In the event of placement of order the following drawings and manuals shall be submitted in six (6) copies to the Chief Engineer, P&C, WBSEDCL, VidyutBhavan (4th floor), Kolkata – 700091 for approval.

- a) General outline dimension drawing of current transformers furnishing front and side elevation, top and bottom plan, views showing all accessories, mounting arrangement on DP structures, including dimension of the bolts, total creepage distance of bushing, electrical diagram for primary and secondary connections with polarity mark, terminal arrangement for secondary terminal box, size of primary terminals, grounding terminals and lifting lugs, quantity of insulating oil, net and shipping weight, shipping dimension etc.

b)Name and rating plate diagram of CT & PT Unit.

13.2 After approval, ten (10) sets of approved drawings and operating and maintenance manual including the instruction manual shall be submitted for our record and distribution to site. (Two complete sets should be directly sent to Chief Engineer, (Dist.), VidyutBhavan (1st floor), Kolkata – 700091) Instruction manual should contain :

- a)A brief description of CT & PT Unit furnishing the constructional features.
- b)Instruction for handling, storing, erection, commissioning and operation and maintenance of CT & PT Unit
- c)General outline drawing of the CT & PT Unit along with all components and accessories.
- d)Marked erection points identifying the component parts of CT & PT Unit.
- e)Detailed dimensions of assembly and description of all accessories.
- f)Detailed views of Core, winding assembly, winding connections and its tappings.
- g)List of spares and other necessary information for CT & PT Unit
- h)A set of approved test certificate.
- i)All notes and legends of the drawings shall be furnished in English and all dimensions shall be marked in metric units.

14. Documents to be submitted at the time of physical delivery at consignee stores :

The following documents to be submitted by the Vendors to the Consignee Stores at the time of physical delivery :-

- a)Copy of Purchase Order.
- b)Copy of Despatch Instruction.
- c)Inspection Test Certificate.
- d)Guarantee Certificate.
- e)Proforma Invoice.
- f)Calculation Sheet for price Variation on the basis of IEEMA as applicable with base date of order.
- g)Seal list and packing list.
- h)Challan in triplicate.
- i)Way bill, if applicable.

15.Credential :Supply & delivery of similar item to Power utility / Govt. organisation within last 3 years need to be submitted within the Bid document.

SCHEDULE-A

SPECIFIC TECHNICAL PARTICULARS OF CURRENT TRANSFORMER

Sl. No.	Description	Rating
i)	Rated system voltage KV (rms)	11
ii)	Highest system voltage KV (rms)	12
iii)	CT Ratio	100-50/ 5A and 50-25/5 A suitable for bulk meter 3 Phase 4 Wire meter, 0.5 S class(Three numbers CT is required)
iv)	Burden (VA)	10 VA
v)	Accuracy Class	0.5 S
vi)	System frequency (Hz)	50
vii)	System neutral earthing	Effectively Earthed
viii)	Installation	Indoor
ix)	Extended current rating	120 %
x)	Rated short time thermal current for 1 second (KA _{rms})	18.4
xi)	Rated Dynamic current KA (peak) (should be at least 2.5 times of	46

	above rating)	
xii)	Rated Insulation level :	
	a) 1.2/50 microsecond impulse withstand voltage (KVpeak)	75
	b) One minute power frequency withstand voltage (KVrms) on primary winding	28
	c) Power frequency withstand voltage for secondary winding for one minute (KVrms)	3
xiii)	Over voltage interturn test	As per IS:2705 & IS 3156
xiv)	Minimum Creepage Distance (Heavily polluted atmosphere)(in mm)	300
xv)	Limit of temp. rise(°C) of windings at rated current	55°C
xvi)	Instrument security factor	Less than 5 for metering core at Lower Ratio
xvii)	Type of Insulation	Class E
xxi)	Painting shed of CT and PT unit	RAL 7032 for external & internal surface
xxii)	Maximum Partial Discharge:	
	a) when energized at 12KV rms	50 PC
	b) when energized at 1.2x12/√3 KV rms	20 PC

SPECIFIC TECHNICAL PARTICULARS OF POTENTIAL TRANSFORMER

<u>Sl. No.</u>	<u>Description</u>	<u>Rating</u>
i)	Rated system voltage KV (rms)	11
ii)	Highest system voltage KV (rms)	12
Iii)	PT Ratio	11000 / √3 /110/√3 V
iv)	Burden (VA)	20 VA
v)	Accuracy Class	0.5
vi)	System frequency (Hz)	50
vii)	Installation	Indoor
viii)	No. of phase/ Connection	Three nos. single phase PT required
ix)	Rated Insulation level:	
	a) 1.2/50 microsecond impulse withstand voltage (KV peak)	75
	b) One minute power frequency withstand voltage (KV rms) on primary winding	28
	c) Power frequency voltage withstand requirements for secondary winding for one minute (KV rms)	3
x)	Limit of temp. rise of winding (°C)	55° C above ambient temperature
xi)	Voltage factor	1.2 cont. & 1.9 for 8 hours
xii)	Maximum Partial Discharge:	
	a) when energized at 12KV rms	50 PC
	b) when energized at 1.2x12/√3 KV rms	20 PC

SPECIFIC TECHNICAL PARTICULARS OF COMMON HOUSING

<u>Sl. No.</u>	<u>Description</u>	<u>Rating</u>
i)	Type of Materials	CRCA
ii)	Thickness of materials	2.5 mm for load bearing & 2.0 mm for non load bearing
Iii)	Degree of Protection	IP 5 X
iv)	Painting Shade	RAL 7032

GUARANTEED TECHNICAL PARTICULARS FOR CURRENT TRANSFORMER
(To be submitted by Bidder)

<u>SI No.</u>	<u>Description</u>	<u>Particulars</u>
1.	Make	
2.	Type	
3.	Reference Standard	
4.	Voltage Grade	
5.	Ratio	
6.	Frequency	
7.	No of core	
8.	Rated VA Burden	
9.	Accuracy Class	
10.	Class of Insulation	
11.	Temperature rise above ambient	
12.	Insulation level-KV (peak/rms)	
13.	a) Short time current rating(KA) for 1.0 sec b) Dynamic current rating(KA peak)	
14.	Instrument security factor	
15.	Magnetizing curve furnished	
16.	Mounting Dimension of CT and PT unit	
17.	Weight of CT&PT Unit	
18.	Material and Diameter of the Primary Stud of the CT & PT unit	
19.	Material and Diameter of the Secondary Stud of the CT & PT unit	
20.	Degree of Protection of enclosure of CT and PT unit	
21.	Painting shed of CT&PT Unit	
22.	Maximum Partial Discharge: a) when energized at 12KV rms b) when energized at $1.2 \times 12 / \sqrt{3}$ KV rms	

GUARANTEED TECHNICAL PARTICULARS FOR POTENTIAL TRANSFORMER
(To be submitted by Bidder)

<u>SI No.</u>	<u>Description</u>	<u>Particulars</u>
1.	Make	
2.	Type	
.	Reference Standard	
4.	Frequency	
5.	Rated Primary Voltage	
6.	Rated Secondary Voltage	
7.	Winding Connection	
8.	Rated VA Burden per phase	
9.	Accuracy Class	
10.	Class of Insulation	
11.	Temperature Rise above Ambient	
12.	Over Voltage Factor	
	a) Continuous	
	b) 8 Hours	
13.	Maximum Partial Discharge	
	a) when energized at 12KV rms	
	b) when energized at $1.2 \times 12 / \sqrt{3}$ KV rms	

SPECIFIC TECHNICAL PARTICULARS OF COMMON HOUSING
(To be submitted by Bidder)

<u>Sl. No.</u>	<u>Description</u>	<u>Rating</u>
i)	Type of Materials	
ii)	Thickness of materials	
Iii)	Degree of Protection	
iv)	Painting Shade	

